

Power

Rate of working and force-speed problems

YOUR AIM TODAY

Connect work, time, force and speed.

YOUR RULE

Read the notes and copy each worked step. Attempt every question before opening page 4. Mark in another colour and repair every error.

Student: Rohan

Start time: _____ Finish time: _____

Date: _____

The ideas you need

Power measures how quickly energy is transferred.

- Power $P=W/t$.
- For a force acting along motion at speed v , $P=Fv$.
- 1 watt=1 joule per second.
- At constant speed, driving force balances resistance.
- Efficiency=useful output power/input power.

MEMORY RULE

Use $P=W/t$ when time is central; use $P=Fv$ when force and speed are central.

Machine

WORKED STEPS

2400 J in 30 s gives $P=80$ W.

Vehicle

WORKED STEPS

At 20 m/s with driving force 600 N, $P=600(20)=12000$ W.

Independent practice

Show every step. Include units in Mechanics.

Question	Working and final answer
1. Find power for 900 J in 15 s.	
2. A 500 N force acts at 12 m/s. Find power.	
3. A 24 kW engine moves at 8 m/s. Find driving force.	
4. At constant speed, resistance is 1500 N at 20 m/s. Find engine power.	
5. A motor input is 5 kW and useful output 4 kW. Find efficiency.	
6. A 2 kW lift raises 100 kg at constant speed. Find speed.	

STOP

Do not continue until all six questions have an attempt.

Mark, diagnose, correct

A wrong answer is useful only when you repair the first wrong step.

Q	Answer and essential working
1	60 W.
2	6000 W=6 kW.
3	$F=P/v=24000/8=3000$ N.
4	$P=1500(20)=30000$ W=30 kW.
5	$4/5=0.8=80\%$.
6	$P=mgv$, so $v=2000/(100g)=2.04$ m/s.

Q	First wrong step	Correct rule

SEND FOR MARKING

Send pages 3 and 4. Ask for a score, the first error in each answer, and one follow-up question per weak skill.